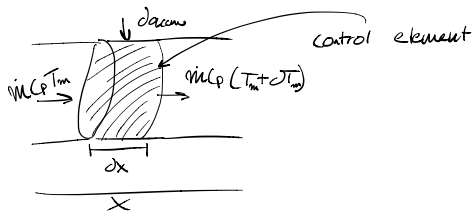


Problem #2



$$\text{flow rate} = 450 \text{ kg/h}$$

$$q' = \alpha x \leftarrow \text{function of length (from entrance)}$$

$$\alpha = \frac{\partial Q}{\partial x}$$

$$C_p \text{ water} = 4.18 \text{ kJ/kg}\cdot\text{K}$$

Rayan Baig
Exam #4

$$A) \dot{m} C_p T_m - \dot{m} C_p T_m - \dot{m} C_p dT_m + dq_{conv} = 0$$

$$dq_{conv} = \dot{m} C_p dT_m$$

$$\rightarrow dq_{conv} = q_{conv} dx = \alpha x dx$$

$$\alpha x dx = \dot{m} C_p dT_m$$

$$\frac{dT_m}{dx} = \frac{\alpha x}{\dot{m} C_p}$$

$$T_m(x) = \frac{\alpha x^2}{2 \dot{m} C_p} + A$$

$$\text{BC: @ } x=0 \rightarrow T_m(0) = 27^\circ\text{C} + 273.15 = 300.15 \text{ K}$$

$$T_m(0) = 0 + A$$

$$300.15 = A$$

$$T_m(x) = \frac{\alpha x^2}{2 \dot{m} C_p} + 300.15 \quad \text{Temp. distribution (in kelvin)}$$

What is the outlet temp of water at 30 meters?

$$T_m(30) = \left(\frac{\partial Q}{\partial x} \right) (30 \text{ m})^2 + 300.15 \text{ K}$$

$$2 \left(\frac{450 \text{ kg/h}}{3600 \text{ s}} \right) \frac{h}{3600 \text{ s}} \left(4.18 \frac{\text{kJ}}{\text{kg}\cdot\text{K}} \right) \frac{1000 \text{ J}}{1 \text{ kJ}} + 300.15 \text{ K}$$

$$T_m(30) = 317.38 \text{ K}$$

Problem #1

$$i) \frac{1}{r} \frac{\partial}{\partial r} \left(D r + \frac{\partial C_A}{\partial r} \right) + k = 0$$

$$\frac{1}{r} \frac{\partial}{\partial r} \left(D r + \frac{\partial C_A}{\partial r} \right) = -k$$

$$\frac{\partial}{\partial r} \left(D r + \frac{\partial C_A}{\partial r} \right) = -k r$$

$$D r + \frac{\partial C_A}{\partial r} = -\frac{k r^2}{2} + C_1$$

$$\frac{\partial C_A}{\partial r} = -\frac{k r}{2 D} + \frac{C_1}{D r}$$

$$C_A(r) = -\frac{k r^2}{6 D^2} - \frac{C_1}{D r} + C_2$$

$$\text{BC: \#1} \rightarrow @ \quad r=0 \quad \frac{\partial C_A}{\partial r} = 0 \quad \text{Assume finite}$$

$$0 = -\frac{k r}{2 D^2} + \frac{C_1}{D r^2} \quad C_1 = 0$$

$$C_A(r) = -\frac{k r^2}{6 D^2} + C_2$$

$$@ \quad r=R \rightarrow C_{AS}$$

$$C_{AS} = -\frac{k R^2}{6 D^2} + C_2$$

$$C_A = C_{AS} + \frac{k R^2}{6 D^2}$$

$$C_A(r) = -\frac{k r^2}{6 D^2} + \frac{k R^2}{6 D^2} + C_{AS}$$

almost!

$$r=R \quad C_{AS}$$

(18/20)

should be -k

(2/20) ii) $\phi_n = \int \frac{k''_{\pi} R^2 C_{A+}^{n+1}}{D_{AB}} = \text{other order } n=0 = \int \frac{k''_{\pi} R^2 D_C R^2}{D_{AB} C_{A+}}$

$\eta = \frac{3}{\phi_n} \left(\frac{2}{n+1} \right)^{1/2} = \frac{3\sqrt{2}}{\int \frac{k''_{\pi} R^2 D_C R^2}{D_{AB} C_{A+}}} \times \eta = 1$

(1/10) iii) effectiveness factor = 1 $\times$

(1/10) iv) When there is NO pore diffusion. $\times$

Bonus:

$-D \frac{\partial C_A}{\partial r} = h_m (C_A - C_{A0})$
@ $r=R$

Another BC? $\therefore$

$C_A(r) = -\frac{k r^2}{6 D_{AB}} - \frac{C_1}{D_{AB} r} + C_2$

$\frac{\partial C_A}{\partial r} = -\frac{k r}{3 D_{AB}} + \frac{C_1}{D_{AB} r^2}$

$D_{AB} \frac{\partial C_A}{\partial r} = -\frac{k r}{3} + \frac{C_1}{r^2}$

$h_m (C_A - C_{A0}) = -\frac{k r}{3} + \frac{C_1}{r^2}$

$C_1 = \left(h_m (C_A - C_{A0}) + \frac{k r}{3} \right) r^2$ @ $r=R$

$C_1 = \left(h_m (C_A - C_{A0}) R^2 + \frac{k R^3}{3} \right)$

$C_1 = \left(h_m (C_A - C_{A0}) R^2 + \frac{k R^3}{3} \right)$

$C_A(r) = -\frac{k r^2}{6 D_{AB}} - \frac{C_1}{D_{AB} r} + C_2$

$C_A(r) = -\frac{k r^2}{6 D_{AB}} - \frac{1}{D_{AB} r} \left(\left(h_m (C_A - C_{A0}) R^2 + \frac{k R^3}{3} \right) \right) + C_2$

@ $r=R$

$C_{A5} = -\frac{k R^2}{6 D_{AB}} - \frac{1}{D_{AB} R} \left(h_m (C_A - C_{A0}) R^2 + \frac{k R^3}{3} \right) + C_2$

$C_2 = C_{A5} + \frac{k R^2}{6 D_{AB}} + \frac{1}{D_{AB} R} \left(h_m (C_A - C_{A0}) R^2 + \frac{k R^3}{3} \right)$

plug back in

Incomplete